



$$f(x) = x^2 - 4ax + a^2$$

$$f(f(x)) = 0 \quad \text{4 pages?}$$

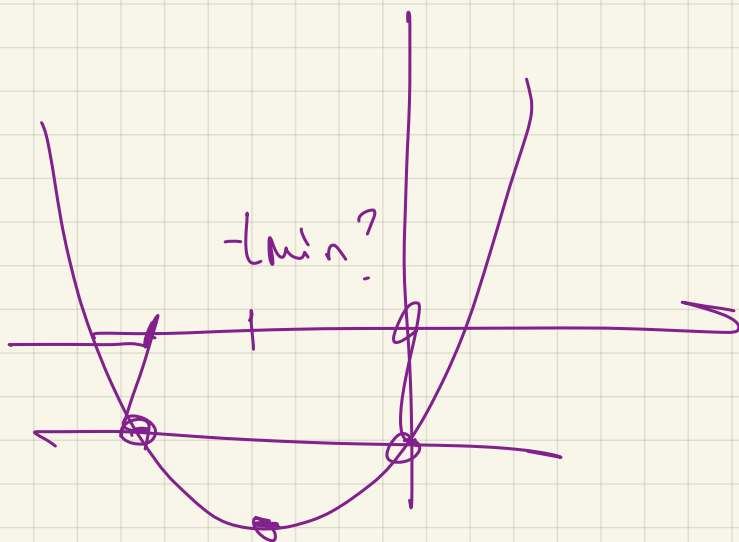
$$f(f(x)) = (x^2 - 4ax + a^2)^2 - 4a(x^2 - 4ax + a^2) + a^2$$

d Bsp?

$$t = f(x)$$

$$x^2 - 4ax + a^2 = t$$

$$f(f(x)) = \underbrace{t^2 - 4at + a^2}_{f(t)} = 0$$



$$t_0 = +\frac{4a}{2} = 2a$$

$$f'(t_0) = 4a^2 - 8a^2 + a^2 = -3a^2$$

$$\frac{4a \pm \sqrt{12a^2}}{2} = 2a \pm \sqrt{3}a$$

$$2a - \sqrt{3}a^2 > -3a^2$$

$$\left\{ \begin{array}{l} 2a + 3a^2 > \sqrt{3}a^2 \\ 2a + 3a^2 \geq 0 \\ 3a^2 \geq 0 \end{array} \right.$$

$$\left\{ \begin{array}{l} (2a + 3a^2)^2 > (3a^2) \\ a(2 + 3a) \geq 0 \\ a \neq 0 \end{array} \right.$$

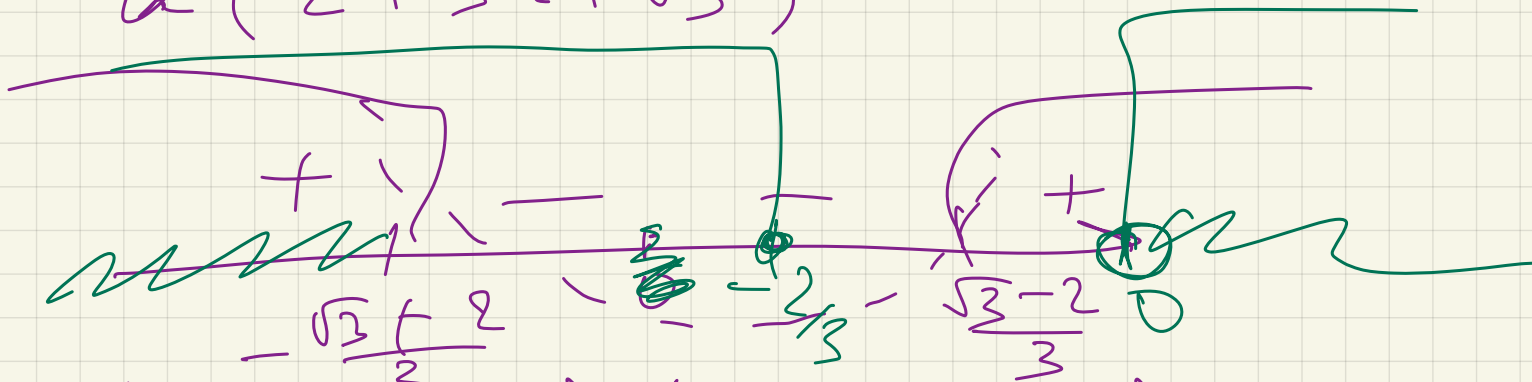
$$\left\{ \begin{array}{l} (2a + 3a^2 - \sqrt{3}a)(2a + 3a^2 + \sqrt{3}a) > 0 \\ \underline{a(2 + 2a) \geq 0} \\ a \neq 0 \end{array} \right. \quad \begin{array}{l} a = -\frac{2}{3} \\ a = 0 \end{array}$$

$$2a + 3a^2 - a\sqrt{3} = 0$$

$$2a + 3a^2 + a\sqrt{3} = 0$$

$$a(2 - \sqrt{3} + 3a) = 0$$

$$a(2 + 3a + \sqrt{3}) = 0$$



$$a^2 (3a + 2 - \sqrt{3})(3a + \sqrt{3} + 2) > 0$$

$$a = 0$$

$$a = \frac{\sqrt{3} - 2}{3}$$

$$a = -\frac{(\sqrt{3} + 2)}{3}$$

$$a \in \left(-\infty; -\frac{(\sqrt{3} + 2)}{3}\right) \cup \left(0; +\infty\right)$$

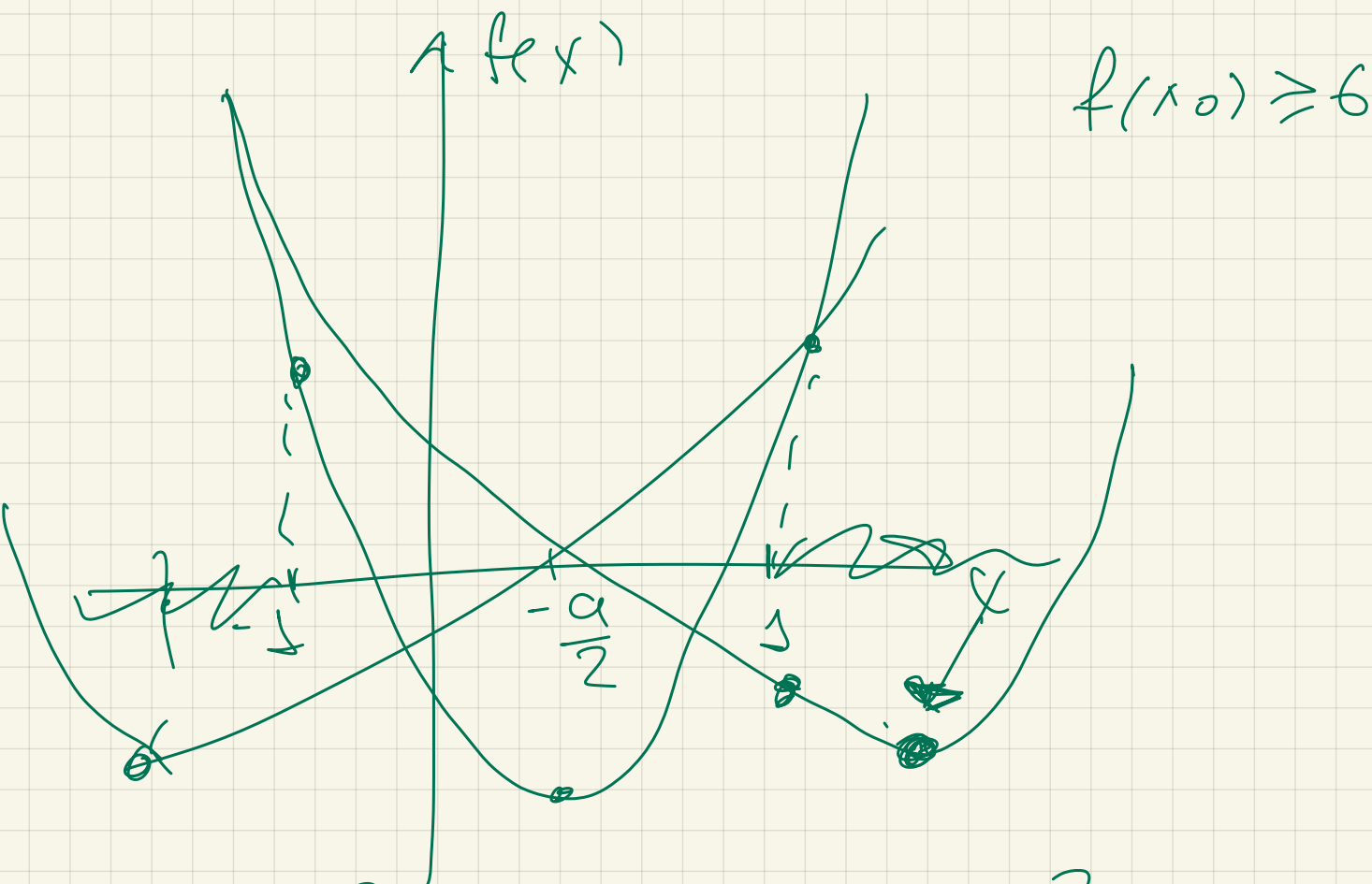
$$4\left(\frac{a^2}{4}\right) - 4a \cdot \frac{a}{2} + a^2 - 2a + 2$$

$$\cancel{a^2} - \cancel{2a^2} + \cancel{a^2} - 2a + 2 = \cancel{(-a^2)} - 2a + 2$$

$$f(x) = 4x^2 + 4ax + a^2 - 2a + 2$$

$$|x| \geq 1$$

we need 3+ we need 6



$$-1 < -\frac{a}{2} < 1$$

$$f(-1) \geq 6 \quad f(1) \geq 6$$

$$\begin{cases} -\frac{a}{2} > 1 \\ f(-\frac{a}{2}) \geq 6 \end{cases}$$

$$-\frac{a}{2} < -1$$

$$f(-\frac{a}{2}) \geq 6$$

$$-1 < -\frac{a}{2} < 1$$

$$f(-1) \geq 0$$

$$f(1) \geq 0$$

$$\left\{ \begin{array}{l} -1 \geq -\frac{a}{2} \geq 1 \\ f(-\frac{a}{2}) \geq 0 \end{array} \right.$$

$$f(-\frac{a}{2}) \geq 0$$

$$-\frac{a}{2} \geq 1$$

$$\frac{a}{2} \leq -1$$

$$a \leq -2$$

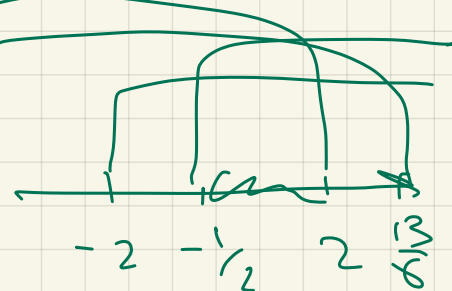
$$-\frac{a}{2} \leq -1$$

$$\frac{a}{2} \geq 1$$

$$-2a + 2 \geq 0$$

$$2a \leq -4$$

$$a \leq -2$$



$$2 \leq a \leq -2$$

$$a \leq -2$$

$$a \geq 2$$

$$a \in [-\infty; -2]$$

$$[-\frac{1}{2}; 2)$$